



2 AN IoT SOLUTION To Keep Tyre Failure At Bay

Investments in transportation are often a necessity for large businesses, educational institutions and organisations. A major concern during travel is the wearing out of tyres in the middle of the road, which can cause unprecedented delays, unwanted hassles and some heavy expenses. Tyre changes for a single commercial vehicle can add up to ₹ 200,000 per year. On top of that, changing tyres on the road can be strenuous. While there are many smart solutions available for businesses to keep a tab on their vehicles' engine health and fuel system, businesses require platforms to stay aware about tyre healths, too. Maharashtra-based startup Tyre Express has taken up this challenge to turn the tables.

Founded in 2015 by Dinesh Wakale and Brijesh Shukla, the B2B startup enables fleet operators to stay aware about the conditions of the tyres by tracking their health and monitor-

ing their performance in real time. Deformation or wearing out of tyres is caused by friction on road, induced dramatically by fluctuations in pressure and temperature, bringing a variable impact on their lifespan.

Tyre pressure depends on the model, size and weight of the vehicle, and also the duration for which the vehicle remains fully-loaded throughout the day. To detect these changes, Tyre Express installs sensors on the walls of the tyres. Sensors can be placed on the inside or the outside, as necessary. These are connected wirelessly to a main hub device, which, in turn, is connected to Microsoft Azure IoT cloud.

When activated, sensors read data like temperature and pressure on the tyres in real time, and send it to the main hub. The hub processes the data and stores it in the cloud. Intelligence on the cloud performs stream analytics and creates graphical visuals and analytical data representations,

which can be referred to by users. Data is stored further in the database. Tyre performance data can be used to recognise any deterioration in tyre health or performance. This, in turn, allows users to make necessary changes in due time to avoid premature tyre failure in the middle of the road. This can improve productivity, costs and on-time performance of business fleets substantially.

At the moment, prevalence of tube-tyres in the market requires sensors to be installed on the outside of the tyres. Tyre Express is working on its design to enable installation of sensors on the inside, which will reduce the chances of loss or damage of said sensors.

The solution is primarily aimed towards businesses and B2B use cases. However, the company is working towards bringing similar solutions for the passenger vertical, too. It is also developing new application features.

—Paromik Chakraborty, technical journalist, EFY

3 A SMART WATER PURIFIER At Zero Purchase Cost

Quality of drinking water is one of those things that we simply cannot afford to compromise—not one bit. Water-borne diseases like diarrhoea, hepatitis, typhoid and cholera, or illnesses due to contaminants like arsenic or lead, can cause large-scale health hazards. Reports suggest that, in India, more than 10,000 lives fell victim to such diseases from 2013 to 2017. To monitor water quality and

reduce contamination, smart sensor-based technologies and filters are being used, which often lead to high capital costs for consumers.

To address this challenge, Bengaluru-based startup OCEO Water, founded by Mahendra Dantewadiya, Rajeev Krishna, Vikram Gulecha and Hasmukh Gulecha, has designed a technology that detects, purifies, measures and monitors drinking

